REVA UNIVERSITY • SCHOOL OF COMPUTING & INFORMATION TECHNOLOGY

AIML-401 • OFFICIAL FACULTY LECTURE MATERIAL

Document Title: Backpropagation & Calculus Foundations Lecture Note

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1. Primary Teaching Points & Lecture Excerpt

“In deep neural networks, backpropagation recursively calculates gradients of the loss function across each hidden layer. However, computing these partial derivatives is fundamentally impossible without applying the calculus chain rule for composite functions. If a student does not understand how composite functions differentiate, backpropagation derivations become confusing.”

2. Mathematical Formulation & Slide Notes

```
dL / dw_jk = (dL / dy_k) * sigma'(z_k) * a_j = delta_k * a_j
delta_j = (sum delta_k * w_jk) * sigma'(z_j)
```

3. Instructor Revision Recommendation

Students must thoroughly master the multivariable chain rule decomposition before attempting backpropagation matrix updates in Assignment 2 and SEE exams. Consult teacher counseling notes for remedial exercises.

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